# THE POWER OF LOGICAL STRUCTURE
# WHAT IS STRUCTURE?

**Michael Smith**
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SLIDEHEROES

What is Structure?

# STRUCTURE: ARRANGING IDEAS & ARGUMENTS

# STRUCTURE V. IDEAS

STRUCTURE ≠ IDEAS

What is Structure?

# THE PRINCIPLE OF ABSTRACTION

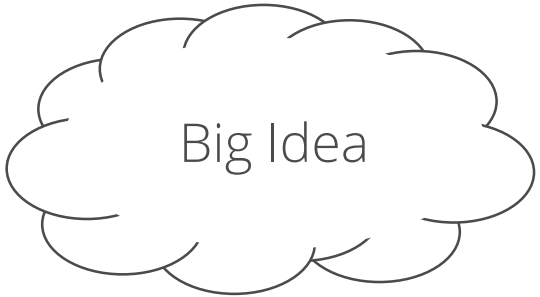
# INFORMAL LOGIC THEORY

1

Big Idea

# INFORMAL LOGIC THEORY

1




2

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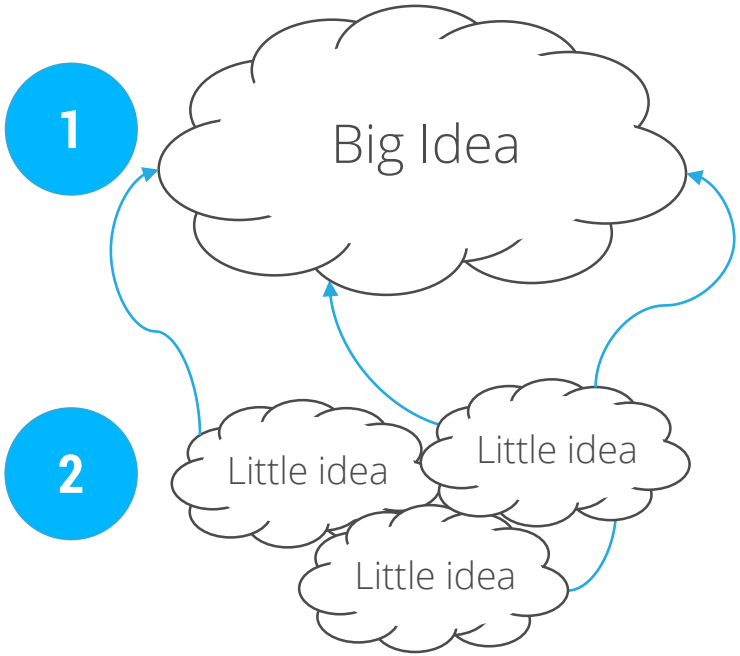

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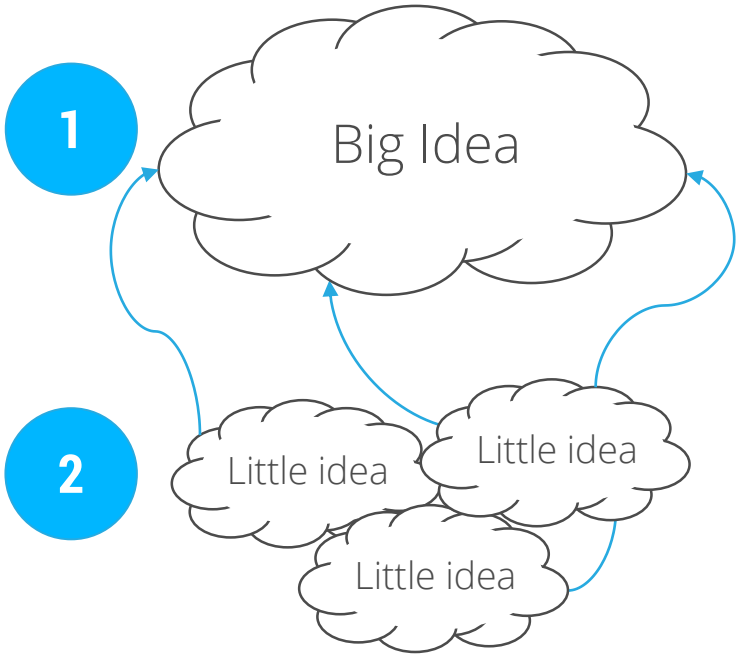
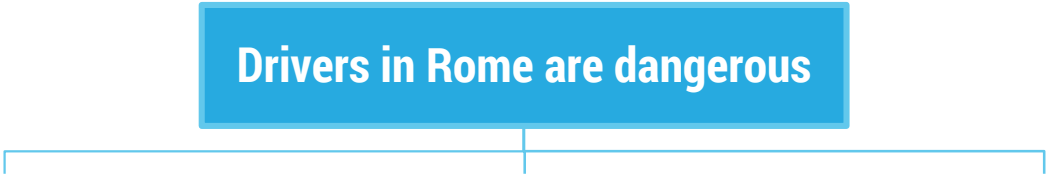



- **Principle of abstraction**
- **Argument map**

# LEVELS OF ABSTRACTION



# LEVELS OF ABSTRACTION

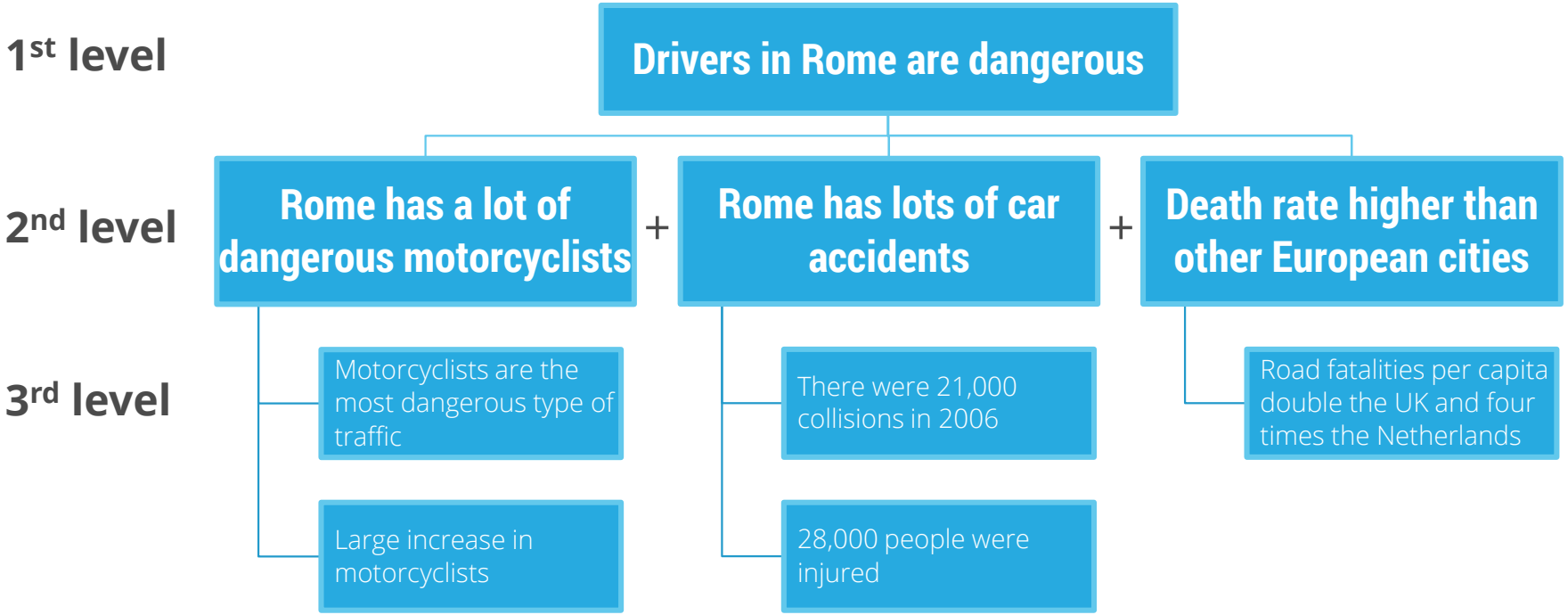
1st level

Drivers in Rome are dangerous

2nd level

Rome has a lot of dangerous motorcyclists + Rome has lots of car accidents + Death rate higher than other European cities

# LEVELS OF ABSTRACTION



What is Structure?

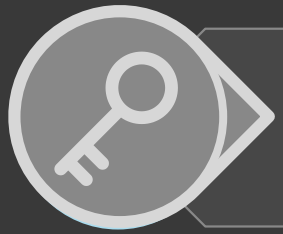
# THE PRINCIPLE OF ABSTRACTION: RULES

# THREE GOLDEN RULES

## EACH LEVEL SUMMARIZES CONCEPTS GROUPED BELOW

- Concepts at any level must be more abstract summaries of the concepts that are grouped below

# THREE GOLDEN RULES



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## IDEAS PRESENTED IN LOGICAL ORDER

- Concepts at an level must be presented in a strict logical order

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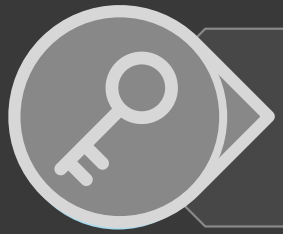

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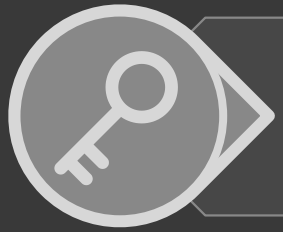

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## GROUPS OF CONCEPTS ALWAYS THE SAME KIND OF IDEA

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# ABSTRACTION RULE #1

# THREE GOLDEN RULES

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# LEVELS OF ABSTRACTION UNIVERSITY EXAMPLE

## ORGANIZATION OF TEACHING FACULTY AT A UNIVERSITY

- Faculty of Science

# LEVELS OF ABSTRACTION UNIVERSITY EXAMPLE

## ORGANIZATION OF TEACHING FACULTY AT A UNIVERSITY

- Faculty of Science
  - Department of Physics

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## ORGANIZATION OF TEACHING FACULTY AT A UNIVERSITY

- Faculty of Science
  - Department of Physics
    - Subject - Physics 101

# LEVELS OF ABSTRACTION UNIVERSITY EXAMPLE

## ORGANIZATION OF TEACHING FACULTY AT A UNIVERSITY

- Faculty of Science
  - Department of Physics
    - Subject - Physics 101
      - ✓ Topic - Fluid dynamics

# LEVELS OF ABSTRACTION UNIVERSITY EXAMPLE

## ORGANIZATION OF TEACHING FACULTY AT A UNIVERSITY

- Faculty of Science
  - Department of Physics
    - Subject - Physics 101
      - ✓ Topic - Fluid dynamics
  - Department of Earth Sciences
  - Department of Biology

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- Faculty of Science
  - Department of Physics
    - Subject - Physics 101
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  - Department of Earth Sciences
  - Department of Biology
- Faculty of Arts
  - Department of History
    - Australian History
      - ✓ 1850-1854 Victorian Gold rush
  - Department of Philosophy
  - Department of Literature
- Faculty of Medicine
  - Department of Immunology
  - Department of Neurosurgery
  - Department of Endocrinology

# INTRODUCTION EXAMPLES

**What should we do?**

CONTEXT

- Revenue growth has stalled, expenses have ballooned and ROE is bad
- Rumours are flying that the CEO and CFO could be on their way out

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**CATALYST**

- Board meeting next month

# INTRODUCTION EXAMPLES

**What should we do?**

**CONTEXT**

- Revenue growth has stalled, expenses have ballooned and ROE is bad
- Rumours are flying that the CEO and CFO could be on their way out

**CATALYST**

- Board meeting next month

**QUESTION**

'How can the financial outlook of Global Universal Bank be improved?'

# GLOBAL UNIVERSAL BANK ARGUMENT MAP

**Global Universal Bank can cut costs by close to $50 million per year**

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**Global Universal Bank can cut costs by close to $50 million per year**

- Streamline central functional activities to save $25MM per year
- Close 25% of rural, low use branches to save $15MM
- Increase branch efficiencies to save $10MM per year
- Outsource IT functions to save $5MM per year

# GLOBAL UNIVERSAL BANK ARGUMENT MAP

**Global Universal Bank can cut costs by close to $50 million per year**

- Streamline central functional activities to save $25MM per year
  - Shift to online training
  - Reduce HR headcount
  - Moratorium on salary increases
- Close 25% of rural, low use branches to save $15MM
  - Close branches with few customers
  - Merge branches close to each other
- Increase branch efficiencies to save $10MM per year
  - Shorten branch opening hours
  - Promote online banking
  - Reduce staffing levels at branches
- Outsource IT functions to save $5MM per year
  - Outsource all back-office IT

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**Global Universal Bank can cut costs by close to $50 million per year**

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**This is a summary of the ideas grouped below**

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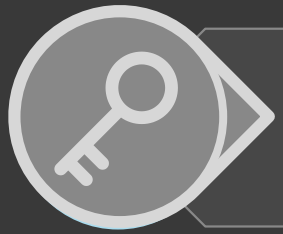
What is Structure?

# ABSTRACTION RULE #2

# THREE GOLDEN RULES



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**Horizontal relationship**

# TWO KEY QUESTIONS

What is the right way of **framing these ideas** to make your argument?

What is **the right order** to introduce your ideas?

What is Structure?

# ABSTRACTION RULE #2: DEDUCTIVE & INDUCTIVE ARGUMENTS

# DEDUCTIVE AND INDUCTIVE REASONING

DEDUCTIVE REASONING

INDUCTIVE REASONING

# DEDUCTIVE AND INDUCTIVE REASONING

**DEDUCTIVE REASONING**

- Starts out with a general statement, or hypothesis, and examines the possibilities to reach a specific, logical conclusion
- Presents ideas in successive steps

**INDUCTIVE REASONING**

# DEDUCTIVE REASONING EXAMPLE

1. John is ill

# DEDUCTIVE REASONING EXAMPLE

1. John is ill
2. If John is ill, he will be unable to attend work

# DEDUCTIVE REASONING EXAMPLE

1. John is ill
2. If John is ill, he will be unable to attend work
3. Therefore, John will be unable to attend work

# DEDUCTIVE AND INDUCTIVE REASONING

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**INDUCTIVE REASONING**

- Also known as induction or 'bottom-up' logic
- Constructs general arguments that are derived from specific examples
- Might conclude with a claim that is only based on a sample of information

# INDUCTIVE REASONING EXAMPLE

- Two independent witnesses claimed John committed the murder

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- Two independent witnesses claimed John committed the murder
- John's fingerprints are the only ones on the murder weapon

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- John confessed to the crime

# INDUCTIVE REASONING EXAMPLE

- Two independent witnesses claimed John committed the murder
- John's fingerprints are the only ones on the murder weapon
- John confessed to the crime
- So, John committed the murder

# GOLDEN RULE

## USE INDUCTIVE ARGUMENTS

Inductive arguments are easier for an audience to absorb because they require less effort to understand

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**This is an inductive argument**

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# ABSTRACTION RULE #2: WAYS TO ORDER IDEAS

# ORDERING IDEAS

## TIME ORDER

What came first, what came second?

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## CAUSE & EFFECT

Cause followed by effect, where the second event (effect) is understood to be a consequence of the first

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Ideas are listed by order of importance

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What came first, what came second?

### CAUSE & EFFECT

Cause followed by effect, where the second event (effect) is understood to be a consequence of the first

### DEGREE / ORDER OF IMPORTANCE

Ideas are listed by order of importance

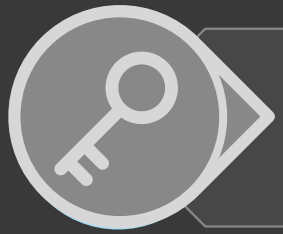
### UNDERLYING PROCESS

A sequence of steps required to accomplish something

What is Structure?

# ABSTRACTION RULE #3

# THREE GOLDEN RULES



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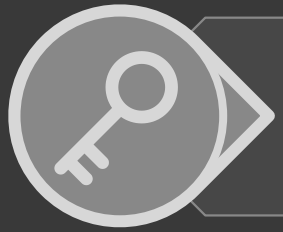

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# THE SAME KIND OF IDEA

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**The Seven Dwarves**

- Doc
- Grumpy
- Happy
- Sleepy
- Bashful
- Sneezy
- Dopey

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## NOT THE SAME KIND OF IDEA

**Pizza toppings? Sizes? Types of crust?**

- Meat
- Thin-crust
- Cheese
- Vegetables
- Sauce
- Extra-large

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**Pizza toppings**

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**Type of crust**

- Thin-crust
- ?
- ?

**Size**

- Extra-large
- ?
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What is Structure?

# ABSTRACTION RULE #3: MECE

# MUTUALLY EXCLUSIVE COLLECTIVELY EXHAUSTIVE

## MUTUALLY EXCLUSIVE (ME)

- Each idea in the list is distinct from each of the other ideas
- No overlap between ideas

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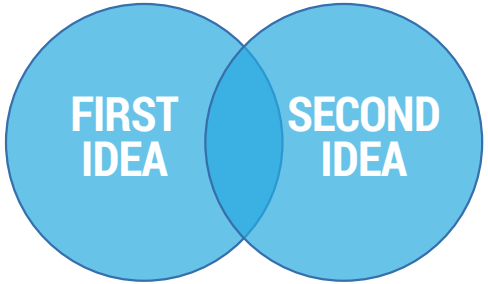

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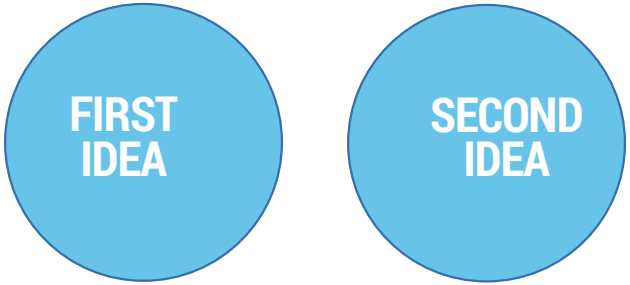

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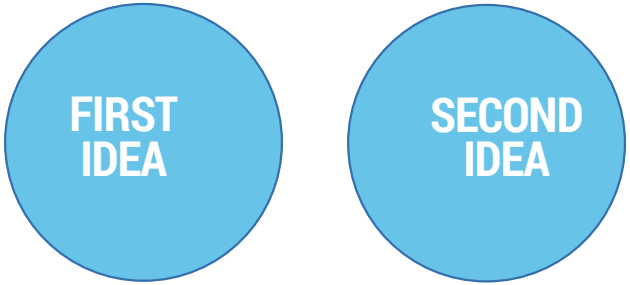
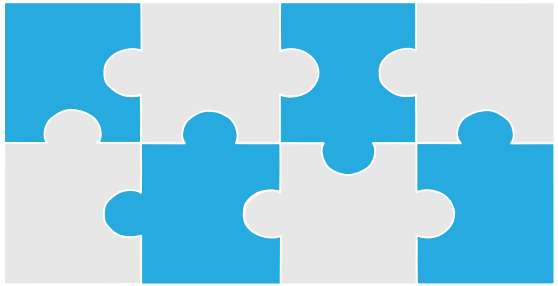



## COLLECTIVELY EXHAUSTIVE (CE)

- Collectively, all the ideas in the list cover all possible components of the idea
- No gaps



What is Structure?

# ABSTRACTION RULE #3: RULE OF 7 (UPDATED)

# TEST OF SHORT TERM MEMORY

## INSTRUCTIONS

- Grab a pen and paper
- Memorize letters when they appear
- AFTER the letters disappear, write down as many letters as you can

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**READY.**

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**READY. SET.**

# TEST OF SHORT TERM MEMORY

## INSTRUCTIONS

- Grab a pen and paper
- Memorize letters when they appear
- AFTER the letters disappear, write down as many letters as you can

**READY. SET. GO!**

U M

**OK. Write down the letters you remember.**

TZLO

**OK. Write down the letters you remember.**

K X C E J O

**OK.  Write down the letters you remember.**

AVCIYSEH

**OK.  Write down the letters you remember.**

L B F Q R M A U X

**OK.  Write down the letters you remember.**

Z Q E C T B U M O N R V

**OK.  Write down the letters you remember.**

# SOPHISTICATION

**OK.  Write down the letters you remember.**

# KEY TAKE AWAY

## NO LISTS > 5

- Salami
- Mozzarella
- Pepperoni
- Mushrooms
- Ground beef
- Cheddar
- Onions
- Green peppers
- Provolone

## CREATE NEW GROUPS

**Meat**

- Salami
- Pepperoni
- Ground beef

**Cheese**

- Mozzarella
- Cheddar
- Provolone

**Vegetables**

- Mushrooms
- Onions
- Green peppers

# LESSON SUMMARY

## GROUP IDEAS INTO HIERARCHIES

Ideas at any level of our presentation must be summaries of the ideas grouped below

## ORDER YOUR IDEAS

Aim to construct inductive-based arguments

## IDEAS IN GROUPS ARE THE SAME KIND

Ideas in the group are always the same kind of idea

## CREATE MECE LISTS

Our groups of ideas should be mutually exclusive and collectively exhaustive

## FOLLOW UPDATED RULE OF 7

At each level of abstraction we should seek to structure our ideas into groups of 4-5 or less